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Design and Implementation of a Crowdsensed Drive Test Mobile Application for Coverage Prediction

COURSE TITLE: Internet Programming & Mobile Programming

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SOFTWARE REQUIREMENTS SPECIFICATION

VERSION 1.0 (SRS)

1. Introduction

1.1 Purpose of the Document

This Software Requirements Specification document presents a detailed description of the functional and non-functional requirements for the proposed Crowdsensed Drive Test Mobile Application for Coverage Prediction. The purpose of this document is to define the expected behaviour, features, constraints, interfaces, and operational characteristics of the system before implementation begins.

This document serves as a communication bridge between developers, project supervisors, testers, system administrators, stakeholders, and future maintainers of the system. It ensures that everyone involved in the project has a common understanding of the objectives and technical requirements of the application.

The SRS document also provides the foundation for software design, implementation, testing, deployment, and maintenance activities. By defining the requirements clearly, the document reduces ambiguity, minimizes project risks, and improves the likelihood of successful system development.

1.2 Scope of the Project

The project focuses on the design and implementation of a mobile crowdsensing application capable of collecting mobile network signal measurements from smartphones and transmitting the collected data to a backend server for processing, analysis, visualization, and coverage prediction.

The system will primarily target Android smartphones because Android provides access to low-level telephony APIs needed for signal monitoring. The application will collect parameters such as RSSI, RSRP, RSRQ, SINR, network type, Cell ID, TAC, and GPS coordinates. These measurements will be aggregated and analyzed in order to generate network coverage maps and identify weak coverage zones across different geographic regions.

The proposed system aims to replace or complement expensive traditional drive testing approaches currently used by mobile network operators in Cameroon. Instead of relying on specialized vehicles and equipment, the system will use ordinary smartphones as passive measurement devices.

The scope of the project includes:

- Mobile application development.
- Background signal monitoring.
- GPS-based geolocation tagging.
- Secure data transmission.
- Offline data buffering.
- Coverage map generation.
- Network analytics.
- Backend server development.
- Database management.

The project excludes direct network optimization by telecommunication operators. The system is designed primarily for monitoring, analysis, and prediction rather than direct control of telecommunications infrastructure.

1.3 Objectives of the System

The main objective of the proposed system is to provide a scalable, low-cost, and efficient solution for mobile network coverage monitoring using crowdsensed smartphone measurements.

Specific objectives include improving visibility into network quality conditions, identifying weak coverage zones, supporting coverage prediction, reducing the cost of traditional drive testing, and providing users with real-time information about network signal quality.

The system also aims to support telecommunication research, improve decision-making for network operators, and contribute to better digital connectivity across Cameroon.

1.4 Problem Statement

Mobile network coverage remains a major challenge in many parts of Cameroon. Existing drive testing methods used by telecommunication operators are expensive, difficult to scale, and incapable of continuously monitoring network conditions across large geographic regions.

Many users experience weak mobile data connectivity, unstable browsing, interrupted calls, poor streaming quality, and inconsistent internet access. Despite these challenges, there is currently no localized crowdsensed network monitoring system specifically designed for the Cameroonian context.

The proposed system addresses this gap by using ordinary smartphones to collect georeferenced signal measurements and generate coverage prediction insights.

1.5 Intended Audience

This document is intended for:

- Software developers.
- System analysts.
- Academic supervisors.
- Test engineers.
- Network engineers.
- System administrators.
- Mobile application designers.
- Project stakeholders.
- Future maintainers of the system.

1.6 Definitions, Acronyms, and Abbreviations

Abbreviation	Definition
SRS	Software Requirements Specification
API	Application Programming Interface
GPS	Global Positioning System
RSSI	Received Signal Strength Indicator
RSRP	Reference Signal Received Power
RSRQ	Reference Signal Received Quality
SINR	Signal-to-Interference-plus-Noise Ratio
LTE	Long Term Evolution
AI – Artificial Intelligence	
UI	User Interface
UX	User Experience
JSON	JavaScript Object Notation

HTTPS	Hypertext Transfer Protocol Secure
SQL	Structured Query Language

1.7 References

References for this project include scholarly journals, mobile crowdsensing research papers, Android telephony documentation, Google Maps API documentation, and studies related to mobile network coverage prediction.

1.8 Document Overview

This document is organized into sections describing the system overview, stakeholder analysis, functional requirements, non-functional requirements, architecture design, database requirements, interface specifications, security requirements, use cases, testing requirements, deployment procedures, maintenance requirements, and future enhancements.

2. Overall Description

2.1 System Perspective

The proposed system consists of two major components: the mobile application and the backend infrastructure.

The mobile application will run on Android smartphones and collect signal measurements together with GPS coordinates. The backend infrastructure will receive, store, process, analyze, and visualize the crowdsensed data.

The system interacts with several external services including Android telephony APIs, GPS services, cloud databases, Google Maps APIs, and analytics tools.

2.2 System Functions

The major system functions include signal monitoring, geolocation tagging, offline storage, secure synchronization, map visualization, analytics generation, user management, operator comparison, and AI-based prediction.

The application will continuously monitor cellular parameters while users move naturally in their environment. Measurements will be tagged with geographic coordinates and synchronized with backend servers.

The backend system will aggregate crowdsensed data from multiple users and generate coverage maps capable of identifying strong and weak signal regions.

2.3 User Characteristics

The primary users of the system include students, smartphone users, network researchers, and telecommunication analysts.

Users may possess varying levels of technical knowledge. Therefore, the application interface must remain simple, intuitive, and easy to understand.

The majority of users are expected to use Android smartphones with active mobile internet connections.

2.4 Operating Environment

The mobile application will operate on Android smartphones running Android 8.0 and above.

The backend infrastructure will operate on cloud servers supporting REST APIs, databases, analytics engines, and web dashboards.

The system will require internet connectivity for synchronization but must also support offline operation.

2.5 Design and Implementation Constraints

The system faces several constraints including:

- Android API restrictions.
- Battery consumption limitations.
- Internet connectivity instability.
- GPS inaccuracies.
- Smartphone hardware differences.
- Privacy regulations.
- Cloud storage limitations.

2.6 Assumptions and Dependencies

The system assumes that users will grant location permissions, maintain active mobile network subscriptions, and periodically connect to the internet for synchronization.

The system also depends on Android telephony APIs, cloud hosting services, and mapping APIs.

3. Stakeholder Analysis

Stakeholder analysis was conducted in order to identify all individuals, groups, organizations, and entities that influence or are influenced by the proposed crowdsensed drive test mobile application for coverage prediction. The analysis helped determine stakeholder expectations, technical concerns, system responsibilities, and regulatory considerations that directly shape the functional and non-functional requirements of the project.

The stakeholders were classified into primary and secondary stakeholders according to their level of interaction with the system and their degree of influence on the project.

3.1 Primary Stakeholders

Primary stakeholders are individuals or groups that directly interact with the proposed system or are directly involved in its development, operation, management, and evaluation.

3.1.1 End Users

End users represent the most important stakeholder group because they directly interact with the application and contribute crowdsensed measurements used for network analysis and coverage prediction. These users include students, smartphone users, researchers, and members of the general public who install the mobile application on their Android smartphones.

The stakeholder validation process confirmed that users expect the application to provide accurate real-time signal monitoring while consuming minimal battery power and internet data. Users also expect the application to operate efficiently in the background without affecting normal smartphone usage.

Stakeholder discussions further revealed that users are concerned about privacy protection, permission management, and secure handling of location information. These concerns directly influenced several non-functional requirements related to security, privacy, usability, and performance optimization.

Users validated the importance of features such as:

- Real-time signal monitoring.
- GPS-based coverage analysis.
- Offline data collection.
- Coverage map visualization.
- Secure synchronization.
- Low battery consumption.
- Minimal internet data usage.

The participation of end users is critical because the effectiveness and accuracy of the crowdsensing system depend heavily on the quantity and quality of crowdsensed measurements contributed by users.

3.1.2 University and Academic Supervisors

The University of Buea together with the project supervisor, Dr. Valery Nkemeni, serve as important academic stakeholders responsible for supervising and validating the project.

During stakeholder validation, the academic stakeholders reviewed the project objectives, methodologies, requirement analysis process, system design approach, and technical feasibility considerations in order to ensure that the project satisfies acceptable software engineering and research standards.

The validation process confirmed that the proposed crowdsensed drive test system addresses a real-world telecommunications problem affecting many mobile network users in Cameroon. The academic stakeholders also validated that the methodologies used for requirement gathering, reverse engineering, requirement analysis, system design, and implementation planning are appropriate for a software engineering research project.

Their recommendations contributed significantly to refining the project scope, improving requirement clarity, strengthening the system architecture, and ensuring alignment with academic research objectives.

The academic stakeholders also emphasized the importance of scalability, maintainability, modular architecture, security, and proper documentation throughout the software development lifecycle.

3.1.3 Backend and System Administrators

The backend developers and system administrators responsible for managing the cloud infrastructure, APIs, databases, analytics engines, and synchronization services are classified as primary stakeholders because they directly manage the technical operation of the system.

The stakeholder validation process confirmed that the backend infrastructure must support large-scale crowdsensed data collection while maintaining high reliability, security, scalability, and performance.

Administrators validated the need for:

- Secure API communication.
- Reliable synchronization services.
- Efficient database management.
- Automated backup systems.

- Real-time monitoring tools.
- Fault tolerance mechanisms.
- Scalable cloud infrastructure.
- Data encryption and authentication.

The validation process also confirmed that backend architecture should remain modular and maintainable in order to support future upgrades such as artificial intelligence prediction systems, advanced analytics, and large-scale geographic expansion.

These requirements directly influenced several non-functional system requirements related to maintainability, reliability, scalability, availability, and security.

3.2 Secondary Stakeholders

Secondary stakeholders are individuals or organizations that may not directly interact with the application daily but are affected by its outputs, analytics, findings, or regulatory implications.

3.2.1 Mobile Network Operators

Mobile network operators such as MTN Cameroon and Orange Cameroon are considered indirect but highly important secondary stakeholders.

The stakeholder validation process confirmed that network operators can benefit significantly from the crowdsensed coverage measurements generated by the application because the collected data can help identify weak coverage zones, optimize antenna placement, improve network planning, and enhance service quality in underserved regions.

The validation activities further confirmed that the selected signal parameters including RSSI, RSRP, RSRQ, SINR, Cell ID, and network type are relevant indicators for network performance analysis and coverage prediction.

The operators' technical standards and measurement practices also influenced the structure of the database schema, analytics framework, and backend data processing mechanisms adopted within the project.

The stakeholder validation process additionally confirmed that crowdsensing provides a more scalable and cost-effective alternative to traditional drive testing methods currently used by telecommunication companies.

3.2.2 Regulatory Bodies

The Agence de Régulation des Télécommunications (ART), Cameroon's telecommunications regulatory authority, is considered a secondary stakeholder because it oversees compliance with telecommunications regulations, data privacy standards, and legal requirements associated with georeferenced data collection.

During stakeholder validation, regulatory concerns related to data protection, user consent management, privacy policies, and secure communication mechanisms were carefully analysed.

The validation process confirmed that the application must:

- Clearly explain user permissions.
- Implement transparent privacy policies.
- Encrypt sensitive data during transmission and storage.
- Anonymize user location information.
- Comply with telecommunications regulations.
- Protect user identity during analytics processing.

The regulatory stakeholder validation therefore significantly influenced the system's security architecture, privacy mechanisms, consent management procedures, and legal compliance requirements.

4. Functional Requirements

4.1 User Registration and Authentication

The system shall allow users to create accounts using email addresses, phone numbers, or third-party authentication services.

The system shall support secure login functionality using encrypted credentials.

The system shall provide password reset functionality.

The system shall maintain authenticated user sessions securely.

4.2 User Permission Management

The system shall request user permissions for location access, background activity, and network monitoring.

The application shall clearly explain why permissions are required.

Users shall be able to revoke permissions at any time.

4.3 Real-Time Signal Monitoring

The application shall collect signal strength parameters including RSSI, RSRP, RSRQ, SINR, Cell ID, and network type.

The system shall update measurements periodically while monitoring is active.

The application shall display real-time signal information to users.

4.4 GPS Location Tracking

The application shall collect GPS coordinates associated with signal measurements.

The system shall support high-accuracy and battery-efficient location tracking modes.

4.5 Background Data Collection

The system shall support automatic background sensing.

The application shall continue collecting measurements while minimized.

The system shall optimize background services to reduce battery usage.

4.6 Offline Data Storage

The application shall temporarily store measurements locally when internet connectivity becomes unavailable.

The system shall synchronize buffered data once connectivity is restored.

4.7 Data Synchronization

The system shall securely transmit collected measurements to backend servers.

The synchronization process shall support retry mechanisms during network failure.

4.8 Coverage Map Visualization

The backend system shall generate coverage maps based on crowdsensed measurements.

Users shall be able to view coverage information geographically.

4.9 Network Analytics Dashboard

The system shall provide dashboards showing signal trends, operator performance, and weak coverage regions.

Administrators shall be able to analyze aggregated network data.

4.10 Network Operator Comparison

The application shall support comparison between operators such as MTN, Orange, and Camtel.

Users shall be able to compare signal quality across geographic regions.

4.11 Notification System

The system shall notify users regarding synchronization status, monitoring activity, and weak signal conditions.

4.12 Data Export and Reporting

The backend shall support exporting reports in CSV and PDF formats.

Administrators shall be able to generate analytics reports.

4.13 Backend Administration

Administrators shall manage users, analytics dashboards, synchronization logs, and system monitoring services.

4.14 User Feedback and Reporting

Users shall be able to report application issues and submit feedback.

4.15 AI-Based Coverage Prediction

The backend shall support machine learning algorithms capable of predicting weak coverage zones.

SUMMARY OF UNBOARDING BY TEAM

Requirement	Priority	Status
The system shall allow users to register an account using an email address and password.	High	Proposed
The system shall support login with registered credentials and maintain a session token.	High	Proposed
The system shall prompt the user to accept required permissions (Location, Phone State) during first launch.	High	Proposed
The system shall explain clearly why each permission is needed before requesting it.	Medium	Proposed
The system shall allow anonymous participation with a device-generated UUID if account creation is declined.	Low	Proposed

5. Non-Functional Requirements

5.1 Performance Requirements

The application shall operate efficiently without causing device freezing or noticeable lag.

Signal collection operations shall complete within acceptable time intervals.

Synchronization operations shall not interrupt user activities.

5.2 Security Requirements

All communications between the mobile application and backend server shall use HTTPS encryption.

Sensitive user data shall be encrypted during storage and transmission.

5.3 Reliability Requirements

The application shall continue operating correctly during temporary network outages.

The backend infrastructure shall support fault tolerance and recovery mechanisms.

5.4 Scalability Requirements

The backend shall support thousands of simultaneous users and large-scale data aggregation.

5.5 Usability Requirements

The user interface shall remain intuitive and easy to navigate.

The system shall support English and French languages.

5.6 Availability Requirements

The backend infrastructure shall maintain high uptime availability.

5.7 Compatibility Requirements

The application shall support Android smartphones with varying hardware specifications.

5.8 Maintainability Requirements

The software architecture shall support future updates, bug fixing, and feature expansion.

5.9 Privacy Requirements

The system shall anonymize sensitive user information.

Users shall be informed regarding data collection policies.

5.10 Portability Requirements

The backend infrastructure shall support migration across cloud providers if necessary.

5.11 Battery Efficiency Requirements

The application shall minimize CPU utilization and GPS polling frequency.

5.12 Data Usage Optimization Requirements

Synchronization shall support compressed data transfer mechanisms.

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6. System Architecture

6.1 High-Level Architecture

The architecture consists of:

- Android mobile application.
- Cloud backend server.
- Database server.
- Analytics engine.
- Web dashboard.
- Mapping services.

6.2 Mobile Application Architecture

The mobile application consists of:

- User Interface Layer.
- Signal Monitoring Module.
- GPS Tracking Module.
- Local Storage Module.
- Synchronization Module.
- Authentication Module.

6.3 Backend Architecture

The backend architecture consists of:

- API Gateway.
- Authentication Services.
- Analytics Engine.
- Coverage Prediction Engine.
- Database Services.
- Reporting Services.

6.4 Database Architecture

The database shall store user information, network measurements, GPS coordinates, analytics data, and logs.

6.5 API Architecture

The backend shall expose RESTful APIs supporting secure communication.

6.6 Cloud Infrastructure

Cloud hosting services shall provide scalability, reliability, backup, and recovery mechanisms.

7. Database Requirements

7.1 User Data

The database shall store usernames, email addresses, encrypted passwords, device identifiers, and preferences.

7.2 Network Measurement Data

The system shall store signal strength measurements, network type information, timestamps, and Cell IDs.

7.3 GPS and Geographic Data

The database shall store latitude, longitude, altitude, and movement speed data.

7.4 Analytics Data

The database shall store aggregated reports and predictive analytics outputs.

7.5 Logs and Monitoring Data

The system shall maintain logs for synchronization events, authentication attempts, and backend monitoring.

8. External Interface Requirements

8.1 User Interface Requirements

The application interface shall support intuitive navigation and responsive design.

Users shall easily access signal monitoring, coverage maps, settings, and analytics screens.

8.2 Hardware Interfaces

The system interfaces with smartphone GPS sensors, cellular modems, storage systems, and internet connectivity hardware.

8.3 Software Interfaces

The system interfaces with Android APIs, Google Maps APIs, cloud services, and database management systems.

8.4 Communication Interfaces

The system shall communicate using HTTPS protocols and REST APIs.

9. System Security Requirements

9.1 Authentication Security

The system shall enforce secure authentication and session management.

9.2 Data Encryption

All sensitive data shall be encrypted during storage and transmission.

9.3 User Privacy Protection

The application shall anonymize sensitive location information before analytics processing.

9.4 Secure API Communication

API endpoints shall require secure authentication tokens.

9.5 Data Backup and Recovery

The backend shall support regular database backup and disaster recovery mechanisms.

10. Use Case Specifications

10.1 Register User

Actor: User

Precondition: User has internet connectivity.

Main Flow: User enters registration details, verifies account, and accesses the application.

Postcondition: User account is successfully created.

10.2 Login User

Actor: User

Precondition: User account exists.

Main Flow: User enters credentials and gains access to the application.

Postcondition: User session becomes active.

10.3 Start Signal Monitoring

Actor: User

Precondition: Permissions granted.

Main Flow: Application collects signal and GPS measurements.

Postcondition: Measurements are stored locally or synchronized.

10.4 Synchronize Data

Actor: System

Precondition: Internet connectivity available.

Main Flow: Buffered data transmitted to backend server.

Postcondition: Backend database updated.

10.5 View Coverage Map

Actor: User

Precondition: Coverage data exists.

Main Flow: User opens map visualization interface.

Postcondition: Coverage map displayed.

10.6 Compare Network Operators

Actor: User

Precondition: Multiple operator datasets available.

Main Flow: User selects operators and locations.

Postcondition: Comparison results displayed.

10.7 Generate Reports

Actor: Administrator

Precondition: Analytics data available.

Main Flow: Administrator generates analytics report.

Postcondition: Report exported successfully.

For this version of the SRS this use cases were draft out, next version will be more explicite.

11. Data Flow Description

11.1 Signal Collection Flow

The smartphone collects telephony measurements through Android APIs and associates them with timestamps.

11.2 GPS Collection Flow

GPS sensors provide geographic coordinates linked to signal measurements.

11.3 Data Synchronization Flow

Buffered data is transmitted securely to backend servers whenever connectivity becomes available.

11.4 Coverage Prediction Flow

Aggregated crowdsensed data is processed by analytics engines and prediction models.

12. System Constraints

The project faces constraints related to Android API limitations, internet instability, smartphone hardware differences, battery consumption, and cloud hosting costs.

13. Assumptions and Risks

The project assumes that users will participate actively in crowdsensed data collection.

Risks include low user adoption, inaccurate GPS measurements, synchronization failures, privacy concerns, and backend scalability challenges.

14. Testing Requirements

The system shall undergo:

- Unit testing.
- Integration testing.
- System testing.
- Performance testing.
- Security testing.
- User acceptance testing.
- Battery consumption testing.
- API testing.

15. Deployment Requirements

The backend infrastructure shall be deployed on scalable cloud hosting services.

The mobile application shall be distributed through APK packages and potentially through the Google Play Store.

16. Maintenance Requirements

The system shall support regular updates, bug fixing, security patching, and feature expansion.

System monitoring tools shall track server performance and synchronization activity.

17. Future Enhancements

Future versions of the system may include:

- Full iOS support.
- Advanced AI prediction models.
- Real-time outage alerts.
- Telecom operator partnerships.
- Route optimization.
- Smart recommendations.
- 5G analytics.
- Crowdsensed QoS analytics.

18. Conclusion

This Software Requirements Specification document provides a comprehensive description of the requirements, architecture, constraints, and operational expectations of the proposed Crowdsensed Drive Test Mobile Application for Coverage Prediction.

The document establishes a strong technical foundation for future design, implementation, testing, deployment, and maintenance activities. By defining the system requirements clearly and comprehensively, the project team can reduce ambiguity, improve development quality, and ensure alignment with stakeholder expectations.

The proposed system represents an important contribution toward scalable and affordable mobile network coverage monitoring in Cameroon.